PAPER ID-410937

Subject Code: HTEH401

Roll No: | | | | | | | | | | | | | |

BTECH-HONS
(SEM IV) THEORY EXAMINATION 2023-24
ELECTRIC & HYBRID VEHICLES

TIME: 3 HRS **M.MARKS: 100**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q no.	Question	Marks	CO
a.	Why do we need hybrid vehicles?	02	1
b.	Write short note on ultra capacitor.	02	1
c.	What is electric drive train.	02	2
d.	Write the type of DC motor.	02	2
e.	What is the difference between sensor and transducer.	02	3
f.	What is BJT.	02	3
g.	Write the component of Propulsion systems.	02	4
h.	What is inverter.	02	4
i.	Define specific power.	02	5
j.	Define the term freeze capacity.	02	5

SECTION B

2. Attempt any *three* of the following: **3 x 10 = 30**

a.	What is the advantage of multiport DC and AC converter based EV charger.	10	1
b.	What is the different control technique used for Switched reluctance motor drive.	10	2
c.	Discuss the role of IOT in the electric vehicle industry.	10	3
d.	Give a brief note on the plug in hybrid vehicle.	10	4
e.	Explain in detail lead-acid battery along with its advantages and disadvantages	10	5

SECTION C

3. Attempt any *one* part of the following: **1 x 10 = 10**

a.	Elucidate the general layout of electric vehicle.	10	1
b.	Explain the difference between ultra-capacitor and battery as an energy storage device for EV.	10	1

4. Attempt any *one* part of the following: **1 x 10 = 10**

a.	Write the construction and principle of induction motor in EV.	10	2
b.	Write note on (i) electrical coupling (ii) Mechanical coupling	10	2

5. Attempt any *one* part of the following: **1 x 10 = 10**

a.	Discuss the construction and operation of IGBT in EV.	10	3
b.	Discuss the various sensor used in self drive cars.	10	3

QP24EP2_311 | 24-Jul-2024 1:36:25 PM | 115.245.67.130

QP24EP2_311 | 24-Jul-2024 1:36:25 PM | 115.245.67.130

PAPER ID-410937

Subject Code: HTEH401

Roll No: | | | | | | | | | | | | |

BTECH-HONS
(SEM IV) THEORY EXAMINATION 2023-24
ELECTRIC & HYBRID VEHICLES

TIME: 3 HRS **M.MARKS: 100**

6. Attempt any *one* part of the following: 1 x 10 = 10

a.	How can regenerate breaking system works in a Hybrid electric vehicle.	10	4
b.	Discuss the Vibration and Noise reduction in Hybrid electric vehicle.	10	4

7. Attempt any *one* part of the following: 1 x 10 = 10

a.	What is a fuel cell ? Describe its basic structure. What are the benefits of a fuel cell?	10	5
b.	Write the characteristics of fuel cell in detail.	10	5

QP24EP2_311 | 24-Jul-2024 1:36:25 PM | 115.245.67.130

QP24EP2_311 | 24-Jul-2024 1:36:25 PM | 115.245.67.130